

The BT136 TRIAC (U1) serves as the main switching component, connecting the load to the 100-250V AC supply via screw terminal U8. MT1 connects to the load via U7, MT2 connects to the 220V live line via U7, and the gate receives the trigger signal. The TRIAC is triggered by a gate current of 10-35mA and switches at the zero-crossing point to reduce EMI.

A snubber circuit (100Ω resistor in series with a 0.1μF capacitor) is connected across MT1 and MT2 to suppress voltage spikes for inductive loads such as motors or fans.

This prevents false triggering and protects the TRIAC from rapid voltage changes.

The load circuit is completed through U7, with MT1 connected to the appliance and MT2 to the AC live supply. A 1A, 250V fuse on the AC input line (via U8) provides overcurrent protection.

An optoisolator bridges the touch sensor and AC switching SSR, ensuring the AC SSR and DC sensor circuit remain electrically isolated yet optically coupled, allowing signal transmission via light without a direct electrical connection.

Optoisolator connection

MOC3041SM (U2) is an optocoupler with zero-crossing detection that provides 5000V isolation between the low-voltage control circuit (TTP223) and the high-voltage solid-state relay (SSR). The Q output pin from the TTP223 connects to pin 1 (anode) of U2 through a 220Ω resistor (R2), and pin 2 (cathode) is connected to GND. Pin 4 (output) of U2 is connected to the gate of the BT136 through a 1kΩ resistor (R1). The datasheet should be referred to for the appropriate R1 selection. Pin 6 of U2 connects to MT2, completing the triggering path.

The optoisolator's internal LED requires approximately 17mA current to activate. This, in turn, triggers the internal phototriac, driving the BT136 gate and ensuring safe, electrically isolated switching. For pinout details and SMD footprint information, refer to the IC datasheet: <https://www.lcsc.com/datasheet/C22373366.pdf>.

For design considerations, R2 is selected to ensure that the LED current remains within the maximum 20mA rating. The connection from pin 6 of the optoisolator

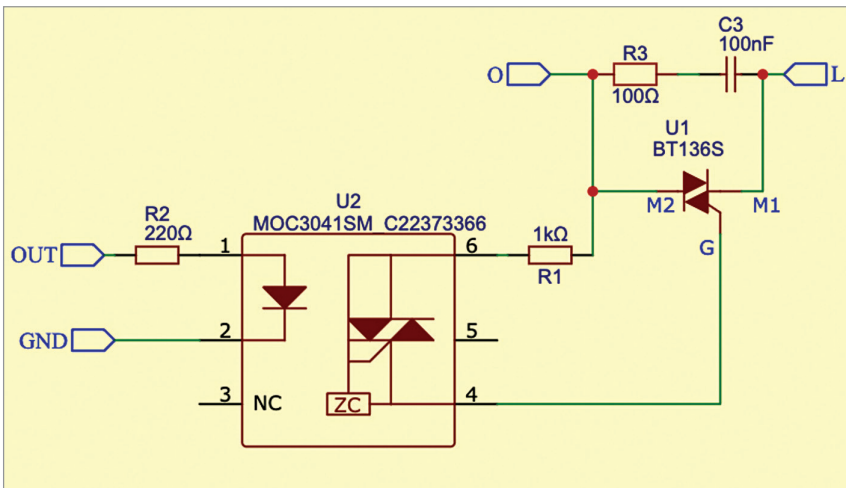


Fig. 5: Solid-state relay circuit (SSR) including snubber circuit (right top)

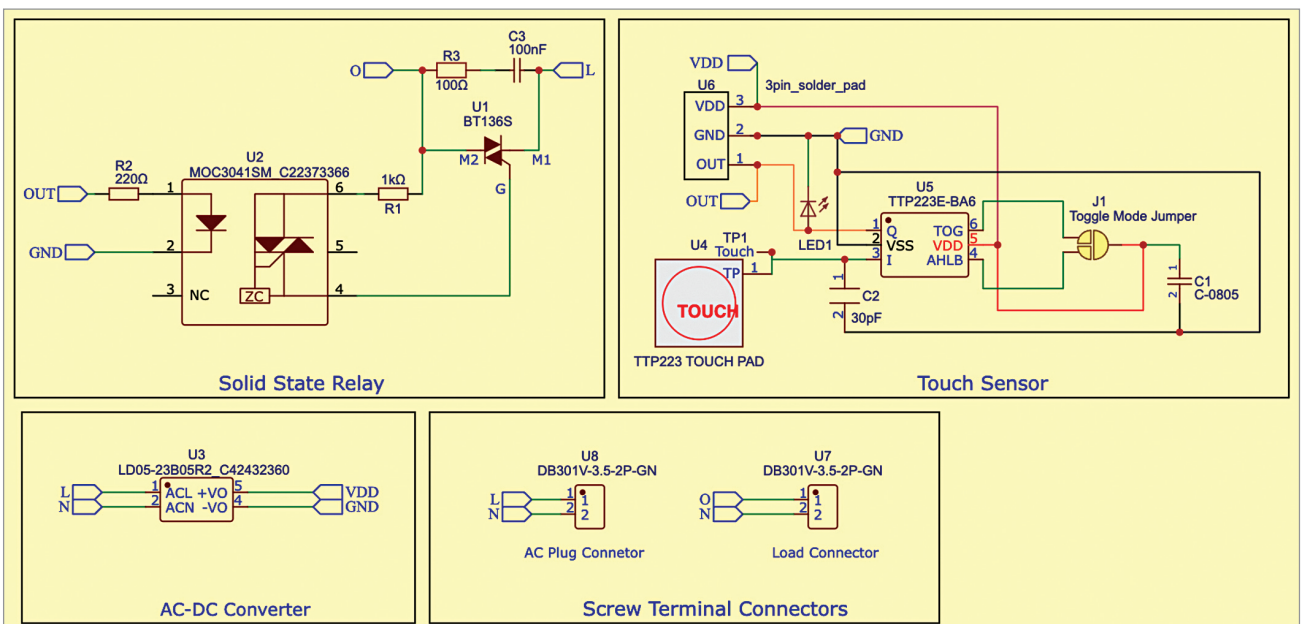


Fig. 6: Complete combined circuit